AE Report for "Optimizing LLM Inference: Fluid-Guided Online Scheduling with Memory Constraints"

This paper proposes a stochastic model for the Large Language Model (LLM) inference optimization problem. Using a fluid approximation as a benchmark, the paper proposes two scheduling algorithms, Wait and Nested Wait, showing that they are asymptotically optimal in terms of throughput, when the arrival rates and processing rates are scaled to infinity.

I have recruited three expert referees for this paper. While there is a consensus that the problem is timely and the research goals are ambitious, there are significant concerns regarding the exposition, the realism of the model assumptions, and the clarity of the theoretical contribution. The key strengths and weaknesses of the paper from the reviewers are summarized as follows.

Strengths

(i) The reviewers unanimously agree that optimizing LLM inference is a high-impact problem of broad interest to both the Operations Research and Computer Science communities.

(ii) Reviewers find the stochastic model and its associated theoretical development interesting and ambitious. One reviewer specifically mentioned that the theoretical development is solid and well-grounded.

Weaknesses

(i) This is the most critical issue. All three reviewers flagged the writing of the paper. The inconsistencies in notation, lack of intuition, and errors make the paper difficult to digest. Two reviewers explicitly stated that the current presentation is "difficult to read" and "prevent a full assessment of the contribution." Another reviewer provided a long list of errors in the paper, many of which are rather careless. They also mentioned that the paper does not provide sufficient intuitions and explanations to the algorithms and analysis.

(ii) Reviewers questioned the realism of the model. Specifically:

a. Reviewers asked for more justifications for Equation (1), which models the iteration time through a simple affine function. One reviewer in particular, provided very helpful discussions on how Equation (1) may or may not capture various features in LLM inference time.

b. There are concerns regarding whether one can design algorithms that never ran into Out-of-Memory (OOM) errors. One reviewer mentioned that OOM could occur due to uncertain number of output lengths, while another reviewer asked whether there should be a memory limit on number of jobs can be held waiting. On a related note, the reviewer also asked for clarification whether the time moving memory (jobs waiting) into the GPU would be time consuming.

(iii) One reviewer raised concern with the theoretical contribution of the paper, commented on that the paper's theoretical results seemed incremental and can be derived through standard stability/positive recurrence results in the stochastic modeling literature. They also mentioned the WAIT and Nested Wait algorithms seem standard. Related to this concern, the reviewer discussed about wanting the paper to explain clearly if a theoretical analysis had to overcome any core difficulties or follows from the standard stochastic literature.

In addition to these key points, all three reviewers have provided various valuable inputs about the papers. I recommend the authors to take careful look and think about all of the points in the reports and how they may help to improve the revision.

My Assessment and Directions for Revision

I agree with the reviewers that the writing quality currently falls short of the journal's standards. However, I believe this research has the potential to make a significant contribution given the importance of the domain. Thus, I am recommending a reject and resubmit, reflecting my assessment that the paper has a plausible, but risky path to publication at OR. Note that because the current exposition (especially in the appendix) obscures the technical details, the reviewers were unable to fully verify the analysis and the practicality of the insights. A clarified revision may reveal technical issues that were previously hidden.

While the referee reports provide valuable pointers, I do not believe it is necessary to incorporate every feature suggested. Here is a list of priorities based on my own assessment:

Complete Expositional Overhaul. Do not simply fix the list of typos identified by the reviewers. The paper requires a coherent rewrite to streamline the narrative and professionalize the mathematical presentation. Here are a few specific things to keep in mind as the paper is revised:

- Currently, the key theoretical results are presented with vague summaries (e.g., "The proof employs the coupling technique…") that are too high-level to be useful. This forces the reader/reviewer to check in the appendix. However, the proofs in appendices are organized more like deposit boxes rather than sections intended for the readers. I encourage the authors to either move some of the proofs in the main text or clearly and explicitly sketch the proof logic in the main text. For those proofs that stay in the appendix, please add some explanations or better, connect the results to standard stochastic literature.

- Related to the previous point, the analysis can be also more efficiently presented by leveraging standard stochastic results instead of from first principles, as one reviewer pointed out. For example, proving basic properties of random walks using Cauchy-Schwarz inequalities (as done in Appendix B.0.1) is unnecessary for the OR audience in stochastic models. Please try to make the connection to established theory in stochastic models and make the paper clearer and easier to verify.
- Finally, I believe the paper can be more efficient when presenting the theoretical results and analysis. For example, a significant portion of Theorem 1 and Proposition 3 is concerned with the knife-edge case where drift is zero. It is well-known in stochastic literature that a random walk with a reflecting boundary scales as $O(1)$ under negative drift and $O(\sqrt{T})$ under zero drift.

Model: Modification or Better Justification. The reviewers have suggested adding various features (e.g., memory overhead for waiting jobs, preemption costs) to make the model realistic. While you should carefully consider these points, I do not require you to complicate the model simply to satisfy every critique. As all modelers know, a simple model with a standard analysis can be very valuable. Thus, you should consider retaining the existing model and persuasively arguing why it is a useful proxy, or alternatively, modifying it to add only the most meaningful features without significantly complicating the analysis. Regardless of your choice, you must be transparent about the limitations of the model. For instance, if you choose not to incorporate hard memory constraints (swapping/OOM), explicitly discuss the real-world scenarios where your assumptions hold and, equally crucially, where they break down.

Clarifying Theoretical Novelty. Currently, the paper lists the development of Asymptotically Optimal Algorithms as one of the key contributions. Please clarify what is the key novelty in waiting and coupling analysis. Also, I acknowledge that properly establishing the connection between a novel domain (LLM inference) and a standard stochastic model is a significant contribution in itself. While the resulting math may appear standard in hindsight, the insight required to structure the problem in this way is often non-obvious ex-ante. If your contribution is the modeling rather than the math, state this clearly.